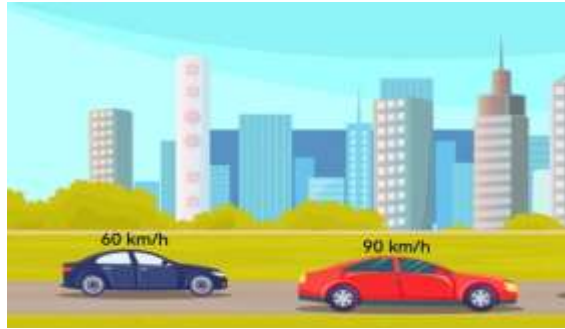


A. Title of card: Indirect Proportion
B. Competency, Element and Performance Criteria: C 2.1: Use numerical concepts in practical situations Element 5: Apply the concepts of rate, ratio and proportion in real life contexts Performance Criteria: Proportion - Level 2: Solve simple problems involving indirect proportion
C. Purpose of activity: This activity will develop an understanding of indirect proportion and apply it in real life contexts.
D. Learning Outcomes: By the end of the lesson, students should be able to: • define indirect proportion • identify real life situations involving indirect proportion • solve simple word problems involving indirect proportion
E. Resources and materials: • Visual aids - pictures of different professions • Simple drawing to colour
F. Teaching trajectories/Implementation guidelines: Introducing Concept of Indirect Proportion Steps: 1. Present students with 2 scenarios: (a) If a painter takes 1 day to paint a house, how long will 2 painters take if they are working at the same rate? (More or less time?) Discuss: "What happens when there are more workers?" (b) If a car travelling at 60 km/h takes 3 hours to reach a destination, how long will it take if the speed is 90 km/h? (More or less time?)

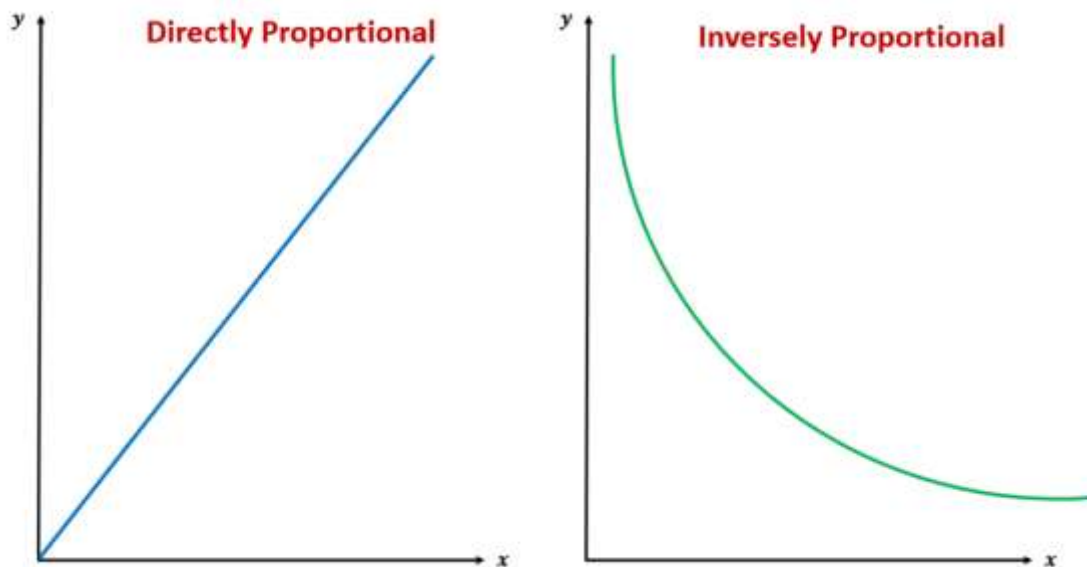


Discuss: "What happens when the speed is increased?"

(c) Explain that this is indirect proportion.

Definition: Two quantities are said to be in indirect proportion or inversely proportional if when one quantity increases, the other quantity decreases.

Study the graph and compare the difference between direct and indirect proportion.



Demonstrating Concept of Indirect Proportion

- (d) Hands on activity and/or use of physical movement to enhance understanding of the concept of indirect proportion.
- Provide students with a simple drawing to colour. Discuss the time taken by 1 student vs time taken by 3 students working together and at the same rate take to finish colouring the same drawing.
 - Have students walk slowly vs quickly across the classroom. Discuss the time taken by each.

Profession-Related Problem Solving.

- (e) Simple word problems are solved as examples. Find below.

Example 1:

A painter takes 4 days to paint a house.

- (a) How long will 2 painters, working at the same rate, take?
 (b) How long will 4 painters, working at the same rate, take?

Number of painters			
Time taken (days)	4	2	1
	1 painter → 4 days	1 painter → 4 days 2 painters → $4 \div 2$ = 2 days	1 painter → 4 days 4 painters → $4 \div 4 = 1$ day

Example 2

2 masons take 6 hours to build a wall. How long will 3 masons, working at the same rate, take?

Number of masons			
Time taken (hours)	12	6	4
	2 masons → 6 hours 1 mason → $6 \times 2 = 12$ hrs	2 masons → 6 hours	2 masons → 6 hours 1 mason → $6 \times 2 = 12$ hrs 3 masons → $12 \div 3 = 4$ hrs

Example 3

3 gardeners take 4 hours to clean the garden. How long will 2 gardeners take?

Number of gardeners			
Time taken (hours)	12	6	4
	3 gardeners → 4 hours 1 gardener → 4×3 = 12 hrs	3 gardeners → 4 hours 1 gardener → 4×3 = 12 hrs 2 gardeners → $12 \div 2$ = 6 hrs	3 gardeners → 4 hours

G. Assessment

Divide students into pairs or groups of 3 to solve the following profession-related problems. Hints are given to struggling students.

Question 1

If 1 dressmaker sews a dress in 8 hours, how long will 2 dressmakers take?

Number of dressmakers		
Time taken (hours)	8	_____
	1 dressmaker → 8 hours	2 dressmakers →

Question 2

A baker prepares all the different cakes for an event in 12 hours; how long will 5 bakers take to prepare all the cakes for the same event? (Answer in both hours and minutes)

1 baker → 12 hours

5 bakers →



Question 3

2 cooks prepare a meal in 3 hours; how long will 3 cooks take to prepare the same meal?

Number of cooks			
Time taken (hours)	_____	3	_____
	2 cooks → 3 hours 1 cook →	2 cooks → 3 hours	2 cooks → 3 hours 1 cook → 3 cooks →

Question 4

4 men take 6 hours to repair a road. How long will it take for 8 men to do the same work?

4 men → 6 hours

1 man →

8 men →



Question 5

If 6 carpenters build a table in 2 days, how long will 3 carpenters take for the same task?

6 carpenters → 2 days

1 carpenter →

3 carpenters →



H. Extension of activity